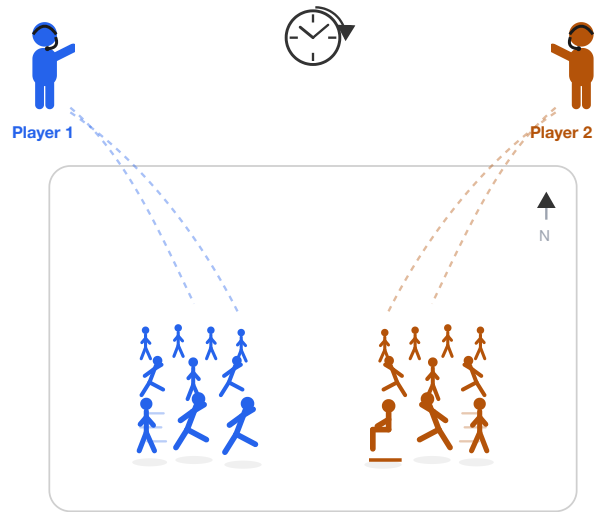


# World Commander

Natural-language command of agent crowds in strategy games, under real-time compute budgets

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This proposal grew from Yubo Huang's interest and insight, under the guidance of Dr. Enmao Diao.



What we build: real-time, language-commanded agent crowds (here, two players commanding their own).

The long-term goal: the next revolutionary video game, played by voice from the commander's chair.

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# 1 Related Work: Efficiency, Untested at Game Speed

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- The interface shipped once (Tom Clancy's EndWar (2008)), but on a 70-word grammar. Language models lifted that limit; they still **cannot act at game pace**.
- LLMs already play RTS through text (TextStarCraft II (NeurIPS 2024)), but only by **pausing the clock**.
- The efficient-inference methods that would fix that (eviction, quantization, distillation, smaller models) are tuned on perplexity, **never in a live game** where dropping the wrong context loses the match.
- A streaming game is the natural stress test, and **nobody has run it** (LLM Game Agents Survey (CSUR 2026) names it a core open challenge).

| **How much does command cost, in latency and memory, at game speed, and how do we drive it down?**

## 2 Preliminaries

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The background this work builds on:

| Area                                 | What it gives us                                                                                                                                                                                  |
|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Reinforcement learning</b>        | A game is a Markov decision process: an agent acts, and win rate is the reward. We need to <i>understand</i> this to read the field; the work runs models by prompting, training none of its own. |
| <b>Efficient LLM inference</b>       | KV-cache eviction (H2O, SnapKV, StreamingLLM, <u>OBCache</u> ), structured pruning, quantization, distillation: the methods this work puts to the test.                                           |
| <b>Vision-language-action models</b> | <u><math>\pi_0</math></u> : a slow vision-language backbone driving a fast action expert, trained by imitation. A model for splitting a slow strategic brain from fast executors.                 |
| <b>The StarCraft II agent stack</b>  | <u>PySC2</u> , <u>TextStarCraft II</u> , LLM-PySC2: the environment we inherit rather than rebuild.                                                                                               |
| <b>Tokenization beyond text</b>      | Byte-pair encoding and <u>graph tokenization</u> (ICLR 2026): the basis for a learned game-state tokenizer.                                                                                       |

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### 3 The Roadmap: Three Projects

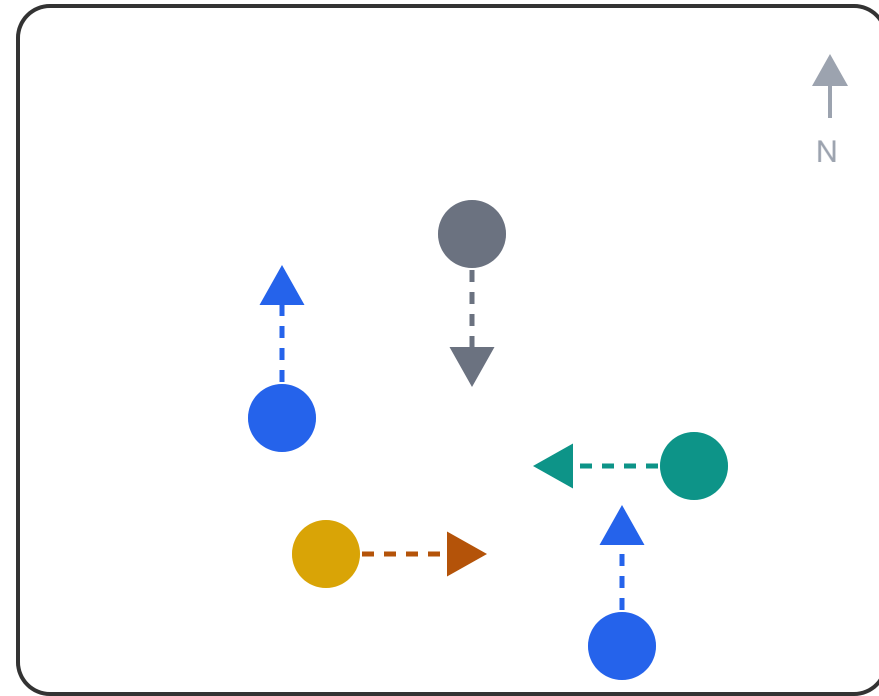
**World Commander** is a research **program**, delivered as three **projects** (each a top-tier paper's worth of work) across environments of growing complexity (a toy room → StarCraft II → multiplayer → a full game), under one shared **harness**. The end-state game is the long-term goal, not a deliverable; each project answers a question its community already cares about, beyond the game.

| Project                                | Phase  | The question                                                  | Who cares                   |
|----------------------------------------|--------|---------------------------------------------------------------|-----------------------------|
| Real-time commander benchmark          | 1      | Which efficiency methods survive a closed-loop game clock?    | KV-cache and pruning        |
| Game-state tokenizer                   | 1 to 2 | Does compact tokenization extend to entity and event streams? | Tokenization beyond text    |
| Crowd motion under budget (embodiment) | 3      | Can language-commanded crowds move in real time on one GPU?   | Motion generation, graphics |

## 4 The Command Arena

**The first step:** a warm-up task that validates the streaming-command infrastructure cheaply, before StarCraft II.

- Colour-tagged agents in a room; each moves in **one of four directions** on command.
- One command is trivial; the test is the **stream**: many, fast.
- Harder with **rate**, **compositional orders** ("everyone but the yellow one"), and **memory** ("the one I moved west").
- Measured: **grounding accuracy**, **command-to-action latency**, **deadline misses**.



**Figure 2: the command arena.** Colour-tagged agents, each moving in one of four directions on command.